Notes: Duffie (JF, 2010)
Asset Price Dynamics with Slow-Moving Capital

Contents

1 Big picture

- **Question.** Why do asset prices often move sharply after a demand or supply shock and then reverse slowly over time?
- **Core idea.** Prices react first to the small amount of risk-bearing capital that is immediately available. Prices then reverse as more investors, dealers, arbitrageurs, or intermediaries become aware of the opportunity, locate counterparties, raise capital, or reallocate balance sheet capacity.
- **Main phrase.** Duffie calls this mechanism **slow-moving capital**: capital does not instantly move to the asset or market where it is most valuable.
- **Main price pattern.** A supply shock produces an immediate price concession, followed by partial or complete reversal. The size of the initial drop and the speed of recovery reveal the frictions that delay capital mobility.
- **Channels.** Slow movement can come from search in over-the-counter markets, limited dealer capital, time needed to raise intermediary capital, price-pressure effects in segmented markets, and limited investor attention.
- **Financial-crisis application.** During 2007–2009, capital constraints at intermediaries generated large deviations from apparent arbitrage relations, such as an extremely negative corporate-bond CDS basis. These deviations reversed as intermediary capital recovered.
- **Model contribution.** The paper presents a stylized general equilibrium model with frequent and infrequent investors. Infrequent investors trade only at fixed intervals. A supply shock is initially absorbed by the small set of investors present in the market and is later spread across the broader investor base.
- **Empirical contribution.** The paper assembles diverse evidence across index deletions, OTC price reactions, CDS basis dislocations, merger-related rebalancing, seasoned equity issues, Treasury issuance, telecom bond issuance, and mutual fund redemptions.
- **Takeaway.** Asset demand curves can slope downward in the short run even when long-run risk premia are modest because the relevant capital is scarce at the moment of the shock.

2 Incremental contribution of the paper

2.1 Contribution relative to classic efficient-market logic

Standard frictionless asset-pricing logic predicts that arbitrage capital should quickly eliminate price deviations from fundamental values. Duffie's contribution is to argue that even highly sophisticated capital is often temporarily immobile. The paper reframes many apparent "anomalies" as equilibrium responses to short-run scarcity of attention, counterparty access, dealer balance sheets, or capital-raising capacity.

Interpretation: The paper does not deny arbitrage. Instead, it emphasizes that arbitrage is a production activity that requires time, balance sheet capacity, search, and attention. Prices can therefore be temporarily distorted even when investors eventually earn returns from absorbing the shock.

2.2 Contribution relative to limits-to-arbitrage research

Prior limits-to-arbitrage theories often emphasize risk, capital constraints, or agency problems. Duffie integrates these ideas into a broader dynamic view: the key variable is not only the amount of arbitrage capital in the economy but also the speed at which it can move to a particular trading opportunity.

- In index deletion events, capital moves over days or weeks.
- In electronic markets, capital can move in seconds or fractions of seconds.

- In catastrophe insurance or dealer-intermediated fixed-income markets, capital may move over months.
- During crises, institutional capital raising and balance sheet repair can take quarters.

Interpretation: The incremental step is to connect short-run price pressure, intermediary constraints, OTC search frictions, and investor inattention under one conceptual mechanism.

2.3 Contribution relative to empirical market microstructure

Microstructure research often studies transitory price impact and order-book dynamics. Duffie's address places similar logic in macro-finance and asset-pricing settings. Price impact and reversal are not only high-frequency phenomena; they also characterize index reconstitutions, bond issuance, credit derivatives, merger arbitrage, and fund redemptions.

Interpretation: The paper expands the scope of price-pressure analysis from trading desks and limit-order books to whole asset markets and financial crises.

2.4 Contribution relative to crisis interpretation

The paper connects crisis-era deviations in arbitrage relations to capital immobility. The extreme negative CDS basis during the crisis is interpreted not as a violation of logic, but as a symptom of constrained dealer and investor balance sheets.

Interpretation: This gives a market-based interpretation of financial-crisis distortions: prices can diverge from textbook relations when the specialized capital needed to enforce those relations is temporarily unavailable.

3 Conceptual framework: slow-moving capital

3.1 The canonical price response

A demand or supply shock changes the desired holdings of an asset. If all investors are immediately attentive and frictionlessly able to trade, the price response should be close to the long-run risk compensation required by the entire investor base. If only a small subset of capital can respond immediately, the initial price movement is much larger.

The typical path is:

1. a shock arrives;
2. immediate liquidity providers absorb it only at a large price concession;
3. additional investors learn, search, raise funds, or rebalance;
4. the asset is gradually transferred to its long-run holders;
5. the price partially or fully reverses.

Interpretation: The speed of reversal is an empirical measure of capital mobility in that market.

3.2 Sources of delay

- **Search costs.** Investors may need time to find counterparties, especially in OTC markets such as bonds, derivatives, securities lending, and reinsurance.
- **Dealer balance sheets.** Dealers can absorb inventory only when they have capital and risk limits available.

- **Capital raising.** Hedge funds, insurers, banks, and dealers need time to raise new capital after losses.
- **Institutional mandates.** Index funds and benchmarked investors may be forced to trade near mechanical dates, while natural buyers arrive later.
- **Limited attention.** Many investors do not monitor every market continuously. They trade periodically or only after becoming aware of an opportunity.

Interpretation: The same price pattern can arise from several frictions. The paper's empirical examples emphasize that the horizon of the reversal helps identify the relevant friction.

4 Empirical motivation and evidence

4.1 Index deletions

When a stock is deleted from the S&P 500, index funds have a strong incentive to sell near the effective deletion date. The natural buyers may not be immediately present, so the price drops sharply before recovering over subsequent weeks.

Interpretation: The index deletion case is clean because the supply shock is largely mechanical and not obviously news about firm fundamentals.

4.2 Limit-order books

In electronic order-book markets, a large sell order consumes bids at successively lower prices. Liquidity-providing investors refresh the book, but not instantaneously. Even when capital moves very fast, the movement is not frictionless at high frequency.

Interpretation: Slow-moving capital is relative to the clock speed of the market. In equity microstructure, "slow" may mean seconds; in reinsurance, it may mean months.

4.3 Catastrophe insurance

After large natural catastrophes, reinsurers' capital is depleted and catastrophe premiums rise sharply. Premiums later decline as capital is raised and redeployed.

Interpretation: This is an example where capital replenishment takes months or years. The price reversal is slower because the relevant capital is specialized and institutionally costly to raise.

4.4 OTC search and securities lending

In OTC markets, buyers and sellers do not necessarily observe a centralized price or instantly find counterparties. Search frictions can cause substantial variation in trade terms and delayed arbitrage.

Interpretation: A supply shock may be absorbed first by those who happen to be connected to the right counterparties, not necessarily by those with the highest long-run valuation.

4.5 Crisis-era CDS basis

The CDS basis is the difference between a CDS rate and the associated par bond yield spread. In frictionless markets, it should be near zero. During the crisis, the basis became highly negative because buying bonds and buying CDS protection required capital, funding, and balance sheet capacity.

Interpretation: The trade was attractive in textbook terms but difficult in institutional terms. The missing input was mobile arbitrage capital.

5 Illustrative model

5.1 Assets and investors

The model has a risk-free bond and a risky asset. The risky asset has a dividend payoff X_t and a stochastic supply Z_t. There are two types of investors:

- **Frequent investors:** they can trade every period.
- **Infrequent investors:** they trade only every k periods.

A fraction q of the investor population is infrequent. Risk preferences are exponential. This keeps individual demand linear in conditional expected excess return divided by conditional variance.

Interpretation: The infrequent investors create delayed capital arrival. They may have capital, but they are not always present to use it.

5.2 State variables

Let D_t denote the demand of the infrequent investors who are trading at time t. Let

$$H_t = (D_{t-1}, D_{t-2}, \dots, D_{t-k+1}) \tag{1}$$

collect the quantities held off market by infrequent investors who traded previously and will not trade again immediately.

The state vector is

$$Y_t = (X_t, Z_t, H_t)'. \tag{2}$$

The equilibrium price is conjectured to be linear:

$$S_t = c'Y_t. \tag{3}$$

Interpretation: The price depends not only on fundamentals and supply, but also on the legacy positions of inattentive investors.

5.3 Demand functions

Given the linear price conjecture, infrequent investor demand and frequent investor demand are linear in the state:

$$D_t = a(c)'Y_t, \tag{4}$$

$$K_t = b(c)'Y_t. \tag{5}$$

The law of motion for the holdings of infrequent investors is

$$H_{t+1,1} = D_t = a(c)'Y_t, \tag{6}$$

$$H_{t+1,i} = H_{t,i-1}, \quad i > 1. \tag{7}$$

Interpretation: Today's trades by infrequent investors become tomorrow's off-market holdings. This creates persistence and slow reversal.

5.4 Market clearing

Market clearing is

$$D_t + K_t = Z_t - \mathbf{1}'H_t. \tag{8}$$

Equivalently, the coefficient vector c must solve the nonlinear fixed point:

$$a(c) + b(c) = g, \tag{9}$$

where $g'Y_t = Z_t - \mathbf{1}'H_t$.

Interpretation: The model is an equilibrium model of price pressure. Price movements are not mechanical; they arise from optimizing demands under infrequent trading.

5.5 Surprise and anticipated shocks

The model studies three types of shocks:

1. **Surprise supply shock:** a large supply shock arrives unexpectedly at date 1.
2. **Anticipated supply shock:** a future supply shock is announced before it occurs.
3. **Imperfectly observed supply shock:** investors initially infer the shock only partially.

Interpretation: The cases map to real markets: index changes, equity issuance, Treasury issuance, and redemption-driven fire sales.

6 Interpretation and identification

6.1 What identifies slow-moving capital

Slow-moving capital is inferred from price paths that show three features:

- an exogenous or quasi-exogenous demand/supply shock;
- a large immediate price impact;
- a subsequent reversal over a horizon related to capital mobility.

Interpretation: The strongest examples are those where the shock is mechanical, such as index deletions or scheduled issuances, because fundamental news is less likely to be the explanation.

6.2 How to separate price pressure from information

A price decline after a sell order could reflect bad news. A slow-moving-capital interpretation is more persuasive when:

- the shock is unrelated to fundamentals;
- the price later reverses;
- the reversal occurs after natural buyers have time to enter;
- similar patterns occur across different assets and institutional settings.

Interpretation: Reversal is the key diagnostic. Pure information shocks should not systematically reverse once capital arrives.

6.3 Limits of the evidence

- The paper is a presidential address and synthesis, not a single structural estimation exercise.
- Different examples involve different frictions, so no one mechanism explains all figures.
- The model is stylized: investors are exogenously inattentive and trade on fixed schedules.
- Some empirical events may combine price pressure with information, financing constraints, and risk-premium changes.

Interpretation: The contribution is a unifying framework and model, not a complete empirical decomposition of every example.

7 Figures

7.1 Figures 1 and 2: initial examples of slow-moving capital

Read:

- Figure 1 shows large negative cumulative returns around S&P 500 deletions followed by gradual recovery.
- Figure 2 shows how OTC search intensity affects the speed and amplitude of price reaction to investor preference shocks.

Economic message: The index deletion example shows price pressure from mechanical selling. The OTC example shows that slower counterparty arrival makes price concessions larger and more persistent.

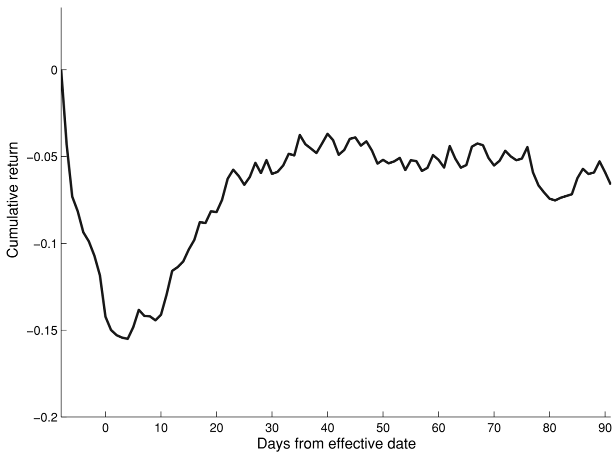

Figure 1. Average cumulative returns for deleted S&P 500 stocks, 1990-2001. The average number of days between the announcement and effective deletion dates is 7.56. The passage of time from announcement to deletion for each equity is re-scaled to 8 days before averaging the cumulative returns during this period across the equities. The original data provided by Jeremy Graveline were augmented by Haoxiang Zhu.

Figure 1: S&P 500 deletions

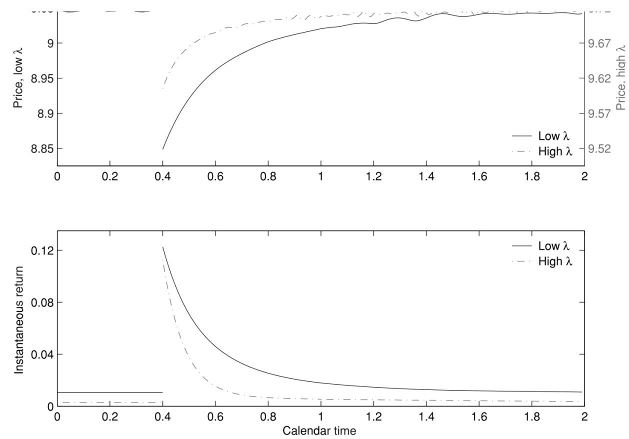

Figure 2. Price reaction to investor preference shocks in an idealized OTC market, with various alternatives for the mean rate λ at which counterparties are located. Source and model details: Duffie, Gârleanu, and Pedersen (2007).

Figure 2: OTC preference shocks

7.2 Figures 3 and 4: capital migration and crisis basis dislocations

Read:

- Figure 3 gives the basic intuition of capital migrating from a low-risk-premium market with excess capital to a high-risk-premium market.
- Figure 4 shows the CDS-bond basis turning strongly negative during the financial crisis.

Economic message: Normal capital migration is slow; crisis-era intermediary capital can become so scarce that apparent arbitrage relations break down for extended periods.

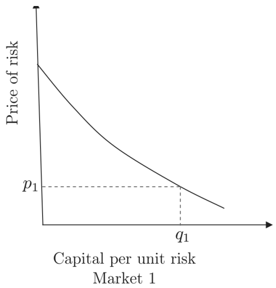

Figure 3. Capital migrates over time from a market with a low risk premium and excess capital to a market with a high risk premium.

Figure 3: capital migration

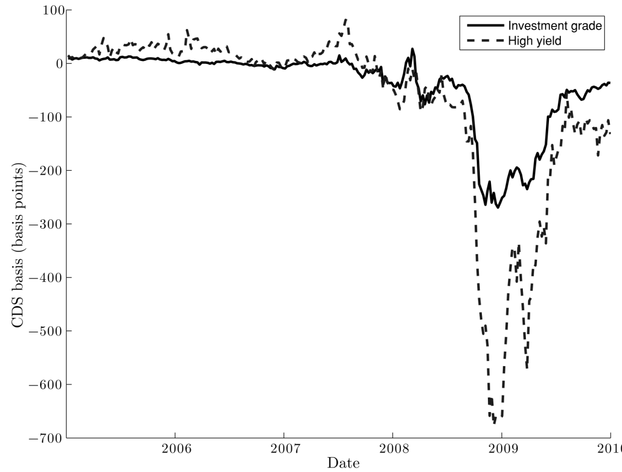

Figure 4. The corporate-bond CDS basis, the difference between the CDS rate and the associated par bond yield spread, is theoretically near zero in frictionless markets. As shown, the average CDS basis across portfolios of U.S. investment-grade bonds and high-yield bonds widened dramatically during the financial crisis and then narrowed as the crisis subsided. The underlying data, kindly provided to the author by Mark Mitchell and Todd Pulvino, cover an average of 484 investment-grades issuers per week and 208 high-yield issuers per week. For additional details, see Mitchell and Pulvino (2010).

Figure 4: CDS basis during the crisis

7.3 Figures 5 and 6: model-implied supply-shock dynamics

Read:

- Figure 5 shows the model price path after a surprise supply shock.
- Figure 6 varies the fraction and trading frequency of inattentive investors.
- Greater investor inattention produces a larger initial impact and slower reversal.

Economic message: The model reproduces the empirical signature of slow-moving capital: sharp initial price pressure followed by gradual recovery.

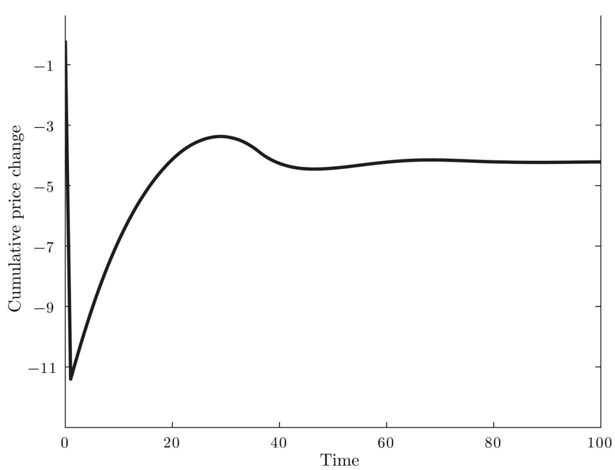

Figure 5. The price path associated with a supply shock on date 1 for the model and parameters described in the text. The parameters are provided on page 1252 of the text.

Figure 5: surprise supply shock in the model

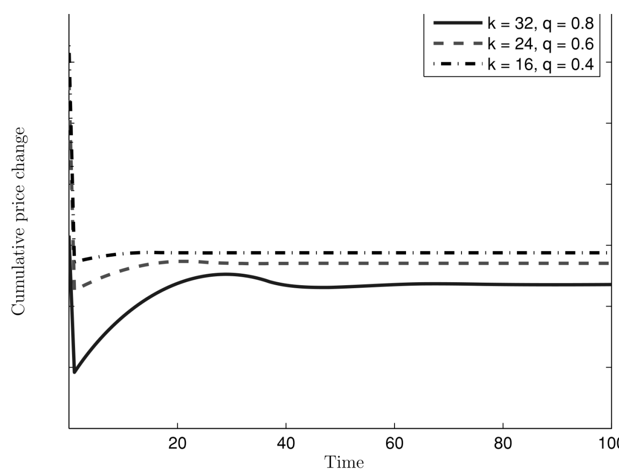

Figure 6. The price paths for the supply shock on date 1 considered in Figure 5, varying the inattentiveness parameters *q* and *k*.

Figure 6: varying q and k

7.4 Figures 7 and 8: merger rebalancing and anticipated supply shocks

Read:

- Figure 7 shows acquirer cumulative returns around mergers, distinguishing index rebalancing cases from no-rebalancing cases.
- Figure 8 shows the model path when a future block sale is announced before it occurs.

Economic message: The model explains why prices can move before the realized trade when investors anticipate forced future supply.

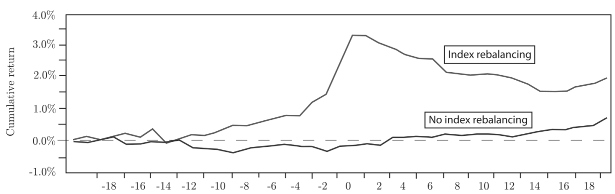

Figure 7. The average of the cumulative stock returns of acquirers, relative to the month of their mergers. Index rebalancing applies if the acquirer is in the S&P 500 and the target is not. Source: Mitchell, Pulvino, and Stafford (2004).

Figure 7: merger-related index rebalancing

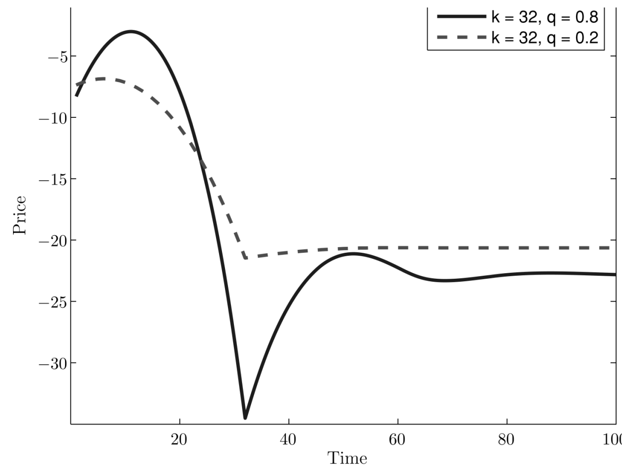

Figure 8. The modeled price path associated with an announcement on date 1 that a block sale occurs on date 32. The model and parameters are described in the text.

Figure 8: anticipated supply shock

7.5 Figures 9 and 10: equity and Treasury issuance

Read:

- Figure 9 shows average price dynamics around secondary equity issuances.
- Figure 10 shows yield elevation around U.S. Treasury issuance, with confidence bands.

Economic message: Even highly liquid securities show issuance-related price pressure when supply must be absorbed by a temporarily limited pool of capital.

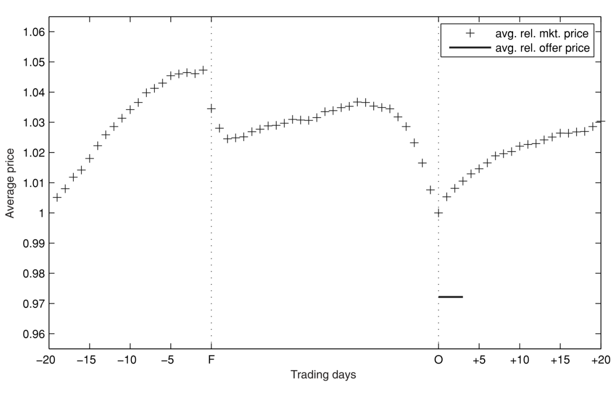

Figure 9. Average price dynamics around secondary equity issuances. The figure, kindly supplied to the author by Jan Peter Kulak, covers 3,850 U.S. industrial firms that undertook a firm-commitment public seasoned offering in the United States between 1986 and 2007. The plotted line shows the average across issuances of the ratio of the secondary market price of the equity to the closing price of the equity on the offering date. Because offerings differ in the number of trading days between the filing announcement and the offering date, the times between filing and offering date are rescaled to the average across the sample of the number of trading days between the filing and the issuance date. Source: Kulak (2008, unpublished data).

Figure 9: secondary equity issuance

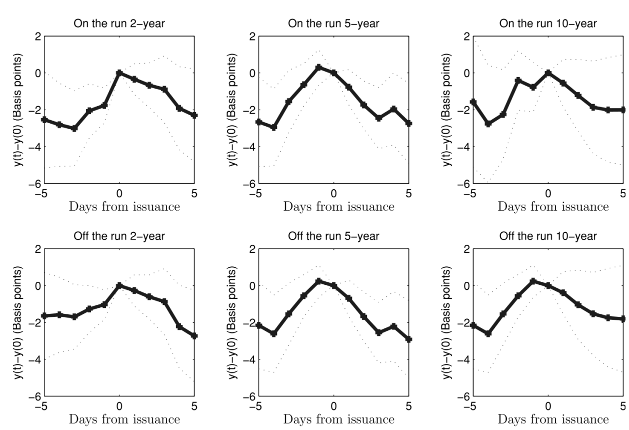

Figure 10. Yield elevation at the issuance of U.S. Treasuries, with 95% "confidence" bands. The figure, kindly provided by Hongjun Yan, covers U.S. Treasury issuances from January 1980 to March 2008. Yields are based on averages of bid and ask prices obtained from the Center for Research in Security Prices. Auction dates are from the U.S. Treasury Department. The sample includes 332 2-year note auctions, 210 5-year note auctions, and 132 10-year note auctions. For each maturity, the differences between the yield on the issuance date and the yield on dates within 5 days of the issuance date are averaged across issuances, for both on-the-run and off-the-run notes. Source: Lou, Yan, and Zang (2010).

Figure 10: Treasury issuance effects

7.6 Figures 11 and 12: spillovers and liquidation pressure

Read:

- Figure 11 shows spillovers in the European telecom debt market after a large Deutsche Telekom issue.
- Figure 12 shows average cumulative returns of equities held by mutual funds experiencing large redemptions.

Economic message: Slow-moving capital can create spillovers across related securities and forced-sale effects across assets held by constrained intermediaries or funds.

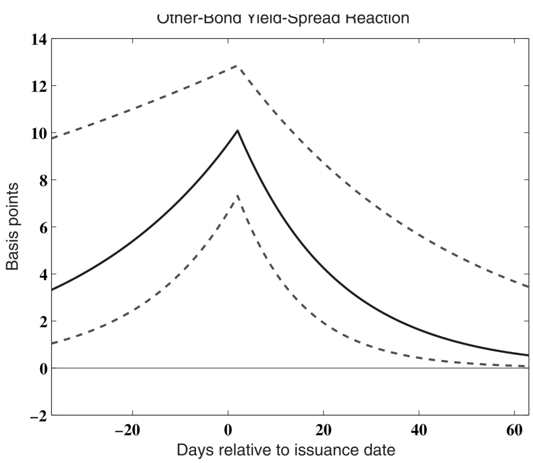

Figure 11. Capital immobility in the Telecom debt market. The estimated impacts on the yields of European Telekom issuers, not including Deutsche Telekom, associated with a particular 16-billion-euro issuance by Deutsche Telekom, using an econometric method explained by Newman and Rierson (2003). Source: Newman and Rierson (2003).

Figure 11: telecom debt market spillovers

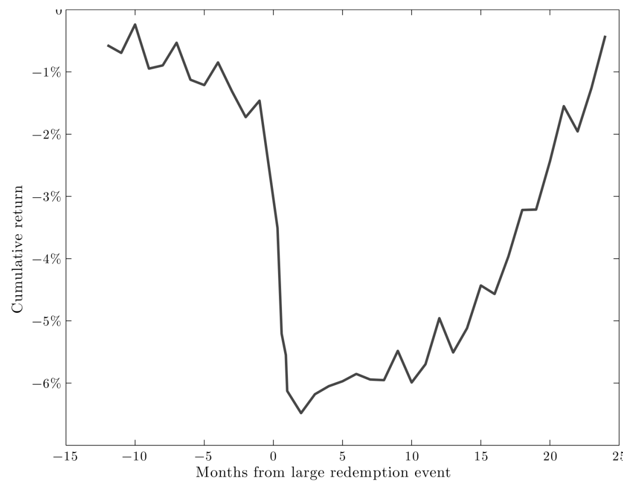

Figure 12. Average cumulative return of equities held by mutual funds experiencing large redemptions, counting months from the date of the large-redemption event. Mutual fund flow data from Thompson Financial were used by Edmans, Goldstein, and Jiang (2009) to calculate net outflows from each equity and each mutual fund. Total outflows across mutual funds, normalized by trading volume, determine "price pressure indices" for each equity, according to a specification stated by Edmans, Goldstein, and Jiang (2009). The cumulative returns plotted are those of equities in the top decile according to price pressure, as of the zero date. Source of cumulative return data: Edmans, Goldstein, and Jiang (2009).

Figure 12: mutual fund redemption pressure

8 Short summary

- Asset prices often react sharply to supply or demand shocks and then partially reverse.
- Duffie interprets this pattern as evidence of slow-moving capital.
- Capital can be slow because investors are inattentive, dealers are constrained, OTC search takes time, or intermediaries must raise capital.
- Crisis-era arbitrage deviations, such as the negative CDS basis, can be understood as temporary scarcity of arbitrage capital.
- The illustrative model has frequent and infrequent investors.
- Infrequent investors trade only every k periods, so only a limited mass of investors absorbs the initial shock.
- The model generates large price impact, reversal, and sometimes overshooting after supply shocks.
- Evidence from index deletions, secondary equity offerings, Treasury issuance, telecom debt issuance, and fund redemptions matches the same qualitative pattern.

9 Conclusion

9.1 Core conclusion

Duffie argues that asset price dynamics often reflect the speed at which investment capital can move to a trading opportunity. The short-run demand curve for an asset can be steep because only a small subset of investors is immediately able or willing to trade. Over time, as more capital arrives, prices revert.

9.2 Economic intuition

Financial markets are not just places where capital is allocated; they are also production systems that require attention, search, balance sheet capacity, and institutional infrastructure. When these inputs are scarce, prices compensate the investors who are immediately available to absorb shocks.

9.3 Takeaway

The paper's broad lesson is that short-run asset pricing requires a theory of capital mobility. Fundamentals and risk premia matter, but the path of prices also depends on who is present, who is constrained, who can find counterparties, and how quickly capital can move.